

1. What is a Constant?

A **constant** is a value that **does not change during the execution of a program**.

Example

```
const int age = 20;
```

Here, **age** is a constant. Its value cannot be changed later in the program.

```
age = 25;    // Error
```

2. Why Do We Use Constants?

Constants are useful when a value should remain the same throughout the program.

Common examples

- Number of days in a week
- Value of π
- Maximum marks
- Tax rate
- University name or fixed code
- Conversion values

Example

```
const int DAYS_IN_WEEK = 7;
```

Using a constant makes the program:

- **Easy to understand**
- **Easy to maintain**
- **Less likely to have accidental changes**
- **More readable**

3. Literal Constants

A **literal constant** is a fixed value written directly in the program.

Example

```
int age = 20;
```

Here, **20** is a **literal constant**.

```
double price = 250.50;  
char grade = 'A';
```

Here:

- **250.50** → literal constant
- **'A'** → literal constant

4. Types of Literal Constants

The common types of literal constants are:

1. **Integer literals**
2. **Floating-point literals**
3. **Character literals**
4. **String literals**
5. **Boolean literals**

4.1 Integer Literals

Integer literals are **whole numbers without a decimal point**.

Examples

```
10  
25  
100  
-50  
0
```

Example:

```
int marks = 100;
```

Here, **100** is an integer literal.

4.2 Floating-Point Literals

Floating-point literals contain a **decimal point**.

Examples

```
10.5  
25.75  
3.14  
-5.5
```

Example:

```
double price = 250.50;
```

Here, **250.50** is a floating-point literal.

4.3 Character Literals

A character literal represents **one character** and is written inside **single quotation marks**.

Examples

```
'A'  
'B'  
'5'  
'$'
```

Example:

```
char grade = 'A';
```

'A' is a character literal.

4.4 String Literals

A string literal is a sequence of characters written inside **double quotation marks**.

Examples

```
"Hello"  
"Pakistan"  
"Computer Science"
```

Example:

```
cout << "Hello World";
```

Here:

```
"Hello World"
```

is a string literal.

4.5 Boolean Literals

Boolean literals have only two values:

```
true  
false
```

Example:

```
bool passed = true;
```

Here, `true` is a Boolean literal.

5. `const` Qualifier

The `const` keyword is used to make a variable **constant**.

Syntax

```
const data_type variable_name = value;
```

Example

```
const int MAX_MARKS = 100;
```

`MAX_MARKS` cannot be changed.

```
MAX_MARKS = 90;    // Error
```

Example Program

```
#include <iostream>
using namespace std;

int main() {

    const int DAYS = 7;

    cout << DAYS;

    return 0;
}
```

Output

```
7
```

Important

A **const** variable should normally be **initialized when it is declared**:

```
const int x = 10;
```

6. #define Directive

#define is a **preprocessor directive** used to define a symbolic constant.

Syntax

```
#define NAME value
```

Example

```
#define PI 3.14
```

Now we can use:

```
cout << PI;
```

Complete Example

```
#include <iostream>
using namespace std;

#define PI 3.14

int main() {

    double radius = 5;
    double area = PI * radius * radius;

    cout << area;

    return 0;
}
```

Output

78.5

7. `const` vs `#define`

<code>const</code>	<code>#define</code>
Uses the <code>const</code> keyword	Uses <code>#define</code>
Has a data type	Does not have a C++ data type
Creates a constant variable	Creates a preprocessor macro
Checked by the compiler	Replaced by the preprocessor
Example: <code>const int X = 10;</code>	Example: <code>#define X 10</code>

Which should beginners prefer?

For defining typed constants in modern C++, **`const` is generally preferred** because it works with C++'s type system.

8. Quick Revision

Constant

A value that does not change during program execution.

```
const int DAYS = 7;
```

Literal Constant

A fixed value written directly in the program.

```
int age = 20;
```

20 is a literal constant.

Common Literal Types

Integer	→ 100
Floating	→ 3.14
Character	→ 'A'
String	→ "Hello"
Boolean	→ true / false

const

```
const int MAX = 100;
```

#define

```
#define PI 3.14
```

★ Easy Rule to Remember

Literal = value written directly

const = constant variable with a type

#define = preprocessor symbolic constant